

addresses as well as the read and write operations. GDDR5 makes use of two write clocks instead of one. One clock called the differential command clock (CK) is for data and addressing and the other is called forwarded differential write clock (WCK). The CK operates at half the WCKs frequency, so if the CK is running at 1.5 GHz the WCK would run at 3 GHz. Each of these two clocks gets a byte of storage each which allows for alignment and synchronisation of data to be stored during the initialisation and training sequences—as explained in the previous paragraph. The data is synchronised on the basis of the value stored in the bytes of each clock. The concept of separating data and operations helps to minimise noise and jitter thereby preserving data integrity at the higher clock speeds that GDDR5 can do.

In conjunction with CDR and dual write clocks, one of the new protocols introduced by GDDR5 is error detection for read/write operations. Note we said *detection*, and not correction. Error correction is not supported and a data re-request occurs if an error is detected. This boosts reliability except for one caveat—as the memory is clocked higher, more errors will occur—and the real world bandwidth may actually suffer as more and more instructions and data need to be re-requested from buffers, in case of an error. However with normal clocks and moderate overclocking this feature should help in maintaining signal integrity and therefore data integrity and system stability.

Hand-in-hand with performance, GDDR5 also has one noteworthy trick that helps in power saving. It's called Address And Data Bus Inversion, and its main function is to reduce the number of binary 0s which are transmitted on both data and address lines. To achieve this, a couple of extra pins are integrated on to a GDDR5 chip—these pins are DBI and ABI pins i.e. data bus inversion and address bus inversion pins, and there are one each of such pins for every byte (8 bits) of data.

In data and address buses, a binary one (1) means a high voltage while a zero (0) indicates a low voltage. In case more than 50 per cent of the bits in a byte are low voltages (zeroes), this technology simply sends an instruction stored in the DBI pin that essentially inverts the binary values for that byte. So, ones become zeroes and zeroes become ones. What this does is ensure that in

any given clock cycle, no block of data, instructions or addresses will ever have more than 50 per cent of its signal containing zeroes. This especially helps in those clock cycles where very little data is being passed through memory—zero (0) values because even in such cases some power consumption still occurs.

And what makes GDDR5 really fast? Very simply GDDR5 doubles the number of DQ links or Data Queue links over predecessors like GDDR3 and GDDR4, so at the same clock speed it's able to (theoretically) get double the bandwidth. So while GDDR3 at 1 GHz had the data effectively clocked at a speed of 2 GHz, GDDR5 runs the same data at an effective speeds of 4 GHz. Note that the core clock speeds of both GDDR3 and GDDR5 are identical—its only data transmission speeds that have changed. So at the same clock speeds 512-bit GDDR3 memory would yield the same throughput as 256-bit GDDR5 memory, which is a tremendous cost saving factor in favour of the newcomer. GDDR5 is also being developed with higher memory capacities in mind—so the density of each memory chip is also higher than GDDR3—this will further simplify the PCB designs of graphics cards that use this new standard as fewer memory data paths need to be designed as fewer chips will offer the same amount of memory. As of now GDDR5 will be expensive to manufacture with only one real outlet for the chips—ATI's Radeon HD4870, but both vendors (NVIDIA and ATI) should skip to GDDR5 over the coming months. The allure of reducing costs, better performance scaling, and lower power consumption will ensure that GDDR3 dies a natural death. It's not a race though—more like a relay, and the baton is about to change hands.

The Future?

We're told that DDR5—the RAM equivalent of GDDR5 is also destined for the desktop sometime in 2011, and its clock speeds could well exceed the fastest processor clock speeds. Obviously quad cores and even hex cores and octal cores would probably be the norm by then. One revolutionary product spawns several others—just like Intel's Core 2 Duo made the DDR2 standard a runaway success as people migrated from their old, slower Athlon 64/DDR systems. Graphics technologies won't stagnate as well, and aside from development on new technologies like GDDR6, we can expect to see a lot more stream processors on the next generation of cards, and memory management techniques will also evolve. The thing to remember is that all this advancement is very necessary and not just a race—software and us, the users, are slowly evolving to these new horizons. ■

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